

# Aavar Khatiwoda

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Falls Church, VA | Active Secret Clearance

## SUMMARY

Versatile software engineer with a wide breadth of experience in diverse technologies, from scripting, systems engineering, and memory compression techniques. Proven track record of leading research teams, spearheading research execution, and upkeeping critical tools.

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, Java, C++, C, Bash, RISC-V Assembly

**Frameworks and Tools:** Linux / Red Hat Enterprise Linux, Virtual Machines, Terminal, Vim, Git, Jenkins, Jira

**Operating Systems & Memory:** Compression and Random-Access Decompression Techniques in Memory

## PROFESSIONAL HISTORY

### Lockheed Martin

Software Engineer

May 2024 - Present

Manassas, VA

- Provide development, implementation, and maintenance of new robust program features, and extend functionality of legacy tools.
- Lead new research project of 10+ engineers to scale unmanned maritime defense over 10+ months. Coordinate project planning and architecture design, validate hardware installs, and develop new software to create and maintain new data transmission pipelines from underwater sensors to secure systems.
- Promote robustness for system maintenance capabilities by developing Bash scripts to capture detailed script log output for key program applications. Provide export procedures/documentation for system logs, allowing developers to quickly detect and fix system failures.

### Northrop Grumman

Software Engineer Intern

Jun 2023 - Aug 2023

Baltimore, MD

- Led team to achieve above-expected story completion targets through effectively coordinating resources and prioritizing high-risk stories as scrum lead.
- Developed new health monitoring scripts used on the Jenkins pipeline to better diagnose and prevent impending build failures, implementing diagnostic filtering of health data on OpenSearch Dashboards.

## RESEARCH

### Virginia Tech, College of Engineering

Undergraduate Research Assistant, CS 4994 - Undergraduate Research

Jan 2024 - May 2024

Blacksburg, VA

- Resolved memory bandwidth limitations of traditional LZ77 and Huffman compression techniques through designing a novel 3-step algorithm tailored to specifications of memory, improving compression and random-access decompression across 4 large and diverse memory dumps.
- Outperformed the DEFLATE algorithm in compression performance across each tested memory dump.

## LEADERSHIP

### CS OpenSource at Virginia Tech, Career Development Resources

Founding Vice President

Jan 2024 - May 2024

Blacksburg, VA

- Established a new career development-focused club targeting students' understanding of AGILE software development.
- Introduced 55+ new students to the software development lifecycle and critical workflow tools [e.g., AGILE, git, terminal].

### Virginia Tech, College of Engineering

Undergraduate Teaching Assistant, CS 1114 - Introduction to Software Design

Aug 2023 - Dec 2023

Blacksburg, VA

- Aided 200+ college students in building a critical understanding of object-oriented programming concepts in Java.

## TECHNICAL PROJECTS

### Software Engineer Portfolio

aavar.studio

2020 - Present

Continuous Learning & Development

- Self-taught ReactJS developer. Built custom professional software engineer portfolio page to highlight relevant technical expertise and leadership/research experience.
- Proficient in UI development with React. Designed fully custom and responsive layouts for common screen sizes up to 4k resolution.

## EDUCATION

### Virginia Tech, College of Engineering

Bachelor of Science in Computer Science

Minor in Mathematics

May 2024

Blacksburg, VA

Dean's List with Distinction